

PRACTICAL COURSE WORKBOOK

Climate Data, Tech & Sustainability

Use evidence to explain a public-interest problem

LEVEL	Beginner
GUIDED TIME	5 hours across 12 lessons
PROJECT	an evidence brief with a reproducible analysis
SKILLS	Climate data, Python, ESG, Carbon, Sustainability tech, Critical evaluation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use Climate data appropriately in a realistic, bounded task.
- Use Python appropriately in a realistic, bounded task.
- Use ESG appropriately in a realistic, bounded task.
- Use Carbon appropriately in a realistic, bounded task.

WHO THIS IS FOR

Public-interest learners using evidence and technology responsibly.

BEFORE YOU BEGIN

- Basic research and spreadsheet skills

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Climate Data, Tech & Sustainability is a structured, practical course covering Climate data, Python, ESG, Carbon, Sustainability tech. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Climate data: Public problems	1. Stakeholders with Climate data 2. Institutions with Python 3. Evidence with ESG
02 Python: Digital methods	1. Open data with Carbon 2. Services with Sustainability tech 3. Participation with Climate data
03 ESG: Policy design	1. Options with Python 2. Impact with ESG 3. Implementation with Carbon
04 Carbon: Accountability	1. Measurement with Sustainability tech 2. Transparency with Climate data 3. Maintenance with Python

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Climate data: Public problems

Apply Climate data through stakeholders, institutions, evidence.

LESSON 01 / 18 MIN

Stakeholders with Climate data

Learning objectives

- Explain stakeholders in the context of Climate Data, Tech & Sustainability.
- Apply Climate data to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

stakeholders, Climate data, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a stakeholders result produced with Climate data?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Institutions with Python

Learning objectives

- Explain institutions in the context of Climate Data, Tech & Sustainability.
- Apply Python to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

institutions, Python, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a institutions result produced with Python?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Evidence with ESG

Learning objectives

- Explain evidence in the context of Climate Data, Tech & Sustainability.
- Apply ESG to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

evidence, ESG, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a evidence result produced with ESG?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Python: Digital methods

Apply Python through open data, services, participation.

LESSON 04 / 30 MIN

Open data with Carbon

Learning objectives

- Explain open data in the context of Climate Data, Tech & Sustainability.
- Apply Carbon to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

open data, Carbon, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a open data result produced with Carbon?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 05 / 18 MIN

Services with Sustainability tech

Learning objectives

- Explain services in the context of Climate Data, Tech & Sustainability.
- Apply Sustainability tech to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

services, Sustainability tech, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a services result produced with Sustainability tech?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 06 / 22 MIN

Participation with Climate data

Learning objectives

- Explain participation in the context of Climate Data, Tech & Sustainability.
- Apply Climate data to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

participation, Climate data, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a participation result produced with Climate data?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

ESG: Policy design

Apply ESG through options, impact, implementation.

LESSON 07 / 26 MIN

Options with Python

Learning objectives

- Explain options in the context of Climate Data, Tech & Sustainability.
- Apply Python to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

options, Python, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a options result produced with Python?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Impact with ESG

Learning objectives

- Explain impact in the context of Climate Data, Tech & Sustainability.
- Apply ESG to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

impact, ESG, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a impact result produced with ESG?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Implementation with Carbon

Learning objectives

- Explain implementation in the context of Climate Data, Tech & Sustainability.
- Apply Carbon to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

implementation, Carbon, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a implementation result produced with Carbon?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Carbon: Accountability

Apply Carbon through measurement, transparency, maintenance.

LESSON 10 / 22 MIN

Measurement with Sustainability tech

Learning objectives

- Explain measurement in the context of Climate Data, Tech & Sustainability.
- Apply Sustainability tech to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

measurement, Sustainability tech, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a measurement result produced with Sustainability tech?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Transparency with Climate data

Learning objectives

- Explain transparency in the context of Climate Data, Tech & Sustainability.
- Apply Climate data to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

transparency, Climate data, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a transparency result produced with Climate data?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Maintenance with Python

Learning objectives

- Explain maintenance in the context of Climate Data, Tech & Sustainability.
- Apply Python to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

maintenance, Python, policy

Retrieval prompt

In Climate Data, Tech & Sustainability, which evidence best supports a maintenance result produced with Python?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Climate Data, Tech & Sustainability project using Climate data, Python, ESG.

DELIVERABLE	A working Climate Data, Tech & Sustainability artefact plus an evidence-based self-review.
TOOLS	A suitable Climate data environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Climate data.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

Open Government Data Toolkit

World Bank - reviewed 2026-08-21

<https://opendatatoolkit.worldbank.org/en/data/opendatatoolkit/home>

Digital government

OECD - reviewed 2026-08-21

<https://www.oecd.org/en/topics/digital-government.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Climate data scope.